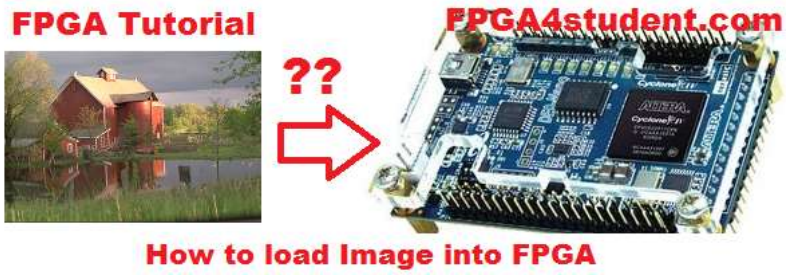
How to load a text file or an image into FPGA

This FPGA tutorial presents two ways to load a text file or an image into FPGA using Verilog or VHDL for image processing. It can be really useful for functional verifications in real-time FPGA image processing projects.

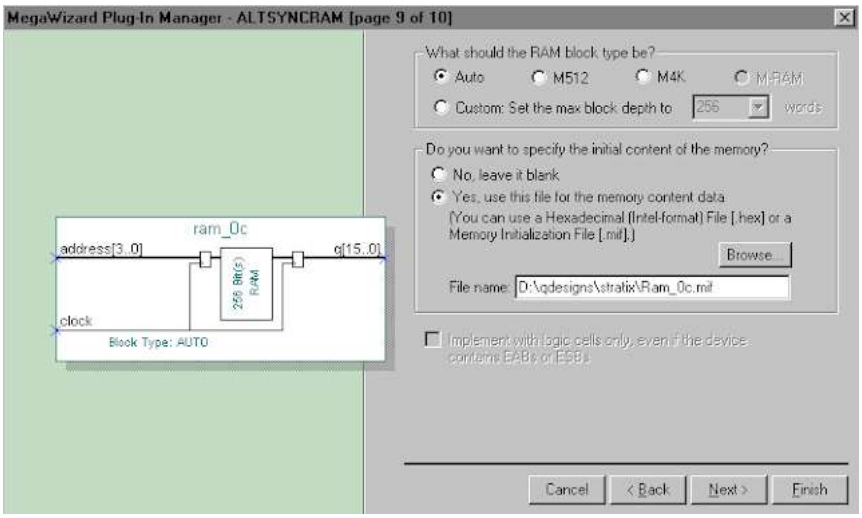


The first way is to load a text file or an image into memory by reading the binary/hexadecimal file using Verilog/ VHDL code.
It is noted that Verilog/VHDL cannot read image file directly, but they can read binary text files. Therefore, the image is required to be converted into binary text files and then use Verilog/VHDL to read it. An image can be easily converted to hexadecimal/ binary data using Matlab, C, Python, etc.

For Verilog code, you can use $readmemh(for hexadecimal data) or $readmemb(for binary data) command to load a converted binary/hexadecimal text file directly. Example Verilog code for loading a text file or an image into FPGA using $readmemh: here.

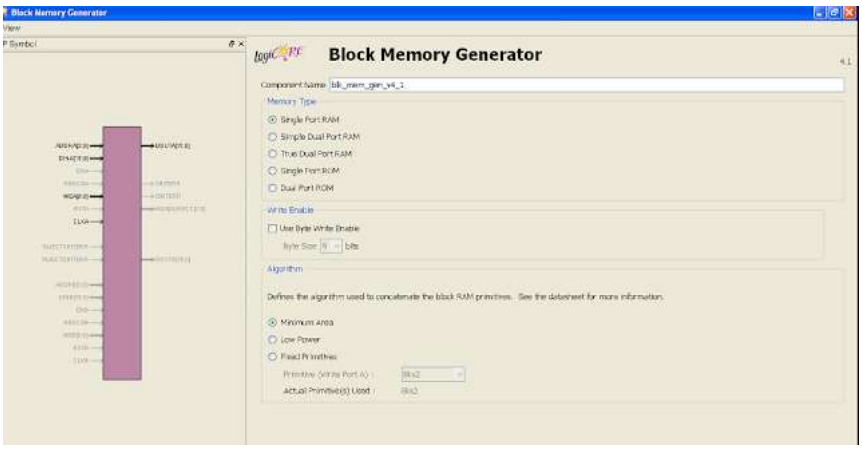
For VHDL code, the TEXTIO package provides necessary commands to read binary text files. Details of the VHDL tutorial on reading image/text file in VHDL can be found here. Another example VHDL code for loading a text file or an image into FPGA: here.

The second way to load a text file or an image file into FPGA is to initialize it as the initial values of the block memory:
1. If you are using Altera FPGA, you can use Mega-Function in the MegaWizard Plug-In Manager in Quartus II to initialize a block memory with the text file or image-converted binary text file in the MIF format. A sample .mif file is here.



You can read more details the process how to use Megafunction in Altera as follows: User guide memory initialization

2. If you are using Xilinx FPGA, use the Core Generator in Xilinx ISE or IP core in Vivado to generate a block memory with the initial content as the text file or image-converted binary text files. The initial data file must be in .coe format. A sample .coe file is here.



You can read more details how to generate a block memory using CORE Generator in Xilinx ISE: Block memory Generator Xilinx

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2. **Basys 3 FPGA OV7670 Camera**
3. **How to load text file or image into FPGA**
4. **Image processing on FPGA using Verilog**
5. **License Plate Recognition on FPGA**
6. **Alarm Clock on FPGA using Verilog**
7. **Digital Clock on FPGA using VHDL**
8. **Simple Verilog code for debouncing buttons on FPGA**
9. **Traffic Light Controller on FPGA**
10. **Car Parking System on FPGA in Verilog**
11. **VHDL code for comparator on FPGA**
12. **Verilog code for Multiplier on FPGA**
13. **N-bit Ring Counter in VHDL on FPGA**
14. **Verilog implementation of Microcontroller on FPGA**
15. **Verilog Carry Look Ahead Multiplier on FPGA**
16. **VHDL Matrix Multiplication on FPGA Xilinx**
17. **Fixed Point Matrix Multiplication on FPGA using Verilog**
18. **Verilog Divider on FPGA**
19. **VHDL code for Microcontroller on FPGA**
20. **VHDL code for FIR Filter on FPGA**
21. **Verilog code for Digital logic components on FPGA**
22. **Delay Timer Implementation on FPGA using Verilog**
23. **Single-Cycle MIPS processor on FPGA using Verilog**
24. **FIFO Verilog Implementation on FPGA**
25. **FIFO VHDL Implementation on FPGA**
26. **Verilog D Flip Flop on FPGA**
27. **Comparator Design on FPGA using Verilog**
28. **D Flip Flop on FPGA using VHDL**
29. **Full Adder Design on FPGA using Verilog**
30. **Full Adder Design on FPGA using VHDL**
31. **Counters on FPGA with Verilog Testbench**
32. **RISC Processor Design on FPGA using Verilog**
33. **Verilog test bench for inout ports on FPGA**
34. **PWM Generator on FPGA using VHDL**
35. **Tic Tac Toe Game on FPGA using Verilog**
36. **VHDL code for ALU on FPGA**
37. **Verilog code for ALU on FPGA**
38. **Counter design on FPGA with VHDL test bench**
39. **Pipelined MIPS Processor on FPGA in Verilog (Part-1)**
40. **Pipelined MIPS Processor on FPGA in Verilog (Part-2)**

41. Pipelined MIPS Processor on FPGA in Verilog (Part-3)
42. Verilog Decoder on FPGA
43. Verilog Multiplexers on FPGA
44. N-bit Adder Design on FPGA in Verilog
45. VHDL ALU on FPGA using N-bit Verilog Adder
46. VHDL Shifter on FPGA
47. Lookup Table VHDL example code on FPGA
48. Coprocessor VHDL Implementation on FPGA
49. Altera FPGA boards for beginners
50. [FPGA Tutorial] Seven-Segment LED Display on Basys 3 FPGA

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